Privacy & AML

BGIN Block#2 (8 March 2021)

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Why do we need Privacy?

- E-commerce didn't take off until HTTPS was available
  - HTTPS encrypts traffic between the web browser and the web server
  - Protects information from anyone snooping the network between the user and the server
- During the 1990s, governments were concerned about the widespread availability of strong encryption
  - Early versions of HTTPS were deliberately hobbled to make them less secure
- Today, strong encryption is mandatory for online banking, making credit card payments, etc.
  - Risks of widespread availability fo encryption are balanced against e-commerce's economic benefits to society
- Today, the development of digital currencies and blockchain/DLT technology is at a similar stage to what the web was in the mid-90s
  - Except it's easier to snoop and datamine because everything's stored permanently on a blockchain

Important factors to consider

- What is the risk of AML / CFT involving privacy coins?
- Is traceability necessary to prevent AML / CFT?
- What are the trade-offs?
- How is blockchain technology likely to develop and evolve?

Objectively Assessing the Risk of Illicit Usage

- Speculation that criminals would adopt privacy coins remains unfounded
- "*Bitcoin is by far the [criminals'] favourite. With our law enforcement customers, we see that over 95% of cases involve Bitcoin.*"
  - Chainalysis Co-founder Jony Levin speaking to Fortune magazine in April 2019
- 51 darknet markets (plus OpenBazaar)
  - Bitcoin: 52 (100%)
  - Ether: 6 (11.5%)
  - Litecoin: 10 (19.2%)
  - Bitcoin Cash: 10 (19.2%)
  - Monero: 13 (25%)
  - Dash: 6 (11.5%)
  - Zcash: 2 (3.8%)

Marketplace	Bitcoin	Ether	Litecoin	Bitcoin Cash	Monero	Zcash	Dash
Open Bazaar*	■	■	■	■	■	■	■
Nightmare	■		■	■	■	■	■
EOS	■	■	■	■			■
DutchDrugz's Psychedelicum	■	■	■	■			■
Silk Road 3.1	■	■	■		■		
BuyBest	■		■	■			■
CStore	■	■	■				
Berlusconi	■		■		■		
Empire	■		■		■		
Dream Market	■			■	■		
24HoursPPC	■			■			
BlackMart	■			■			
People Drug Store	■			■			
DeDope German Weed Store	■			■			
Mollyworld No 1 provider of Crystal or Pills	■				■		
Cannazon	■				■		
Wall Street	■				■		
Tochka	■	■					
Core	■				■		
Cryptonia	■				■		
Unicc	■						■
Namaste LSD	■				■		
SilPP	■		■				
Apollon	■				■		
TorShops	■						
Guns Dark Market	■						
Counterfeiting Center	■						
Gold & Diamonds	■						
Football Money	■						
EuroGuns	■						
USfakeIDs	■						
Global Dreams	■						
Onion Pharma	■						
QualityKing	■						
FlexxCocaine NO1 Cocain Provider 100% pure	■						
HapLapHQDrugsShop	■						
US Pharmacy	■						
BitPharma	■						
Smokeables! Finest Organic Cannabis, shipping from the USA	■						
EuCanna First class Cannabis HealthCare	■						
Ecviano Crew Luxury Weed Wholesale	■						
The French Connection	■						
DeDope	■						
Brainmagic	■						
CannabisUK	■						
ChemSpain	■						
CharlieUK	■						
Hydra	■						
Majestic Garden	■						
CGMC	■						
Thclear	■						
Heineken Cards	■						
	100%	11.5%	19.2%	19.2%	25%	3.8%	11.5%

Objectively Assessing the Risk of Illicit Usage

Marketplace	Bitcoin	Ether	Litecoin	Bitcoin Cash	Monero	Zcash	Dash
Nightmare							

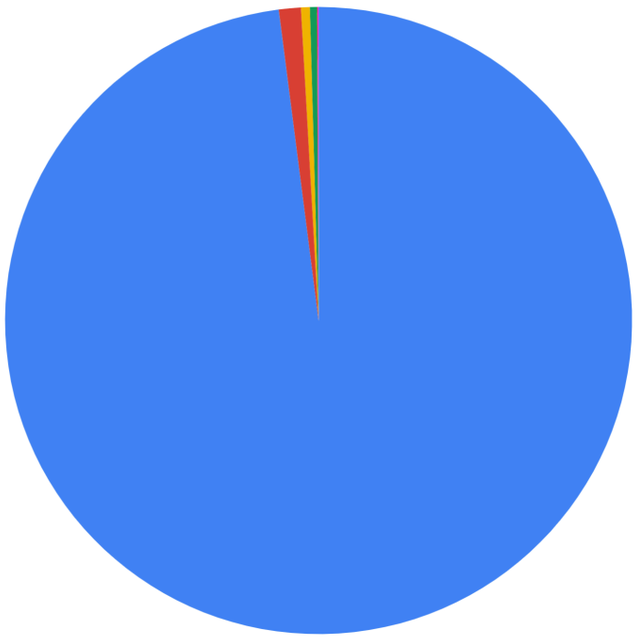
Data leaked from the Nightmare darknet market by a hacker in July 2019 indicates that, even when criminals have the option of using privacy coins, most stick with Bitcoin.

Market total sales

22721245.6 USD
2294.68223779 BTC
3243.70881813 XMR
1171.40183899 LTC
280.981902 BCH
61.49032092 DASH
169.75573555 ZEC

% of sales by coin:

- Bitcoin 97.97%
- Monero 1.13%
- Litecoin 0.46%
- Bitcoin Cash 0.36%
- Zcash 0.05%
- Dash 0.03%
- All privacy coins 1.21%



Is Traceability Necessary for AML / CFT ?

- Well-regulated exchanges (e.g. Gemini, Coinbase) currently list Zcash while remaining compliant with their AML / CFT obligations
- In September 2020, Perkins Coie published a whitepaper titled *Anti-Money Laundering Regulation of Privacy-Enabling Cryptocurrencies.*

Conclusion

Privacy coins reflect a nascent, but important, effort to safeguard our fundamental interest in personal and commercial financial privacy. The AML risks of privacy coins, while real, do not require specific, tailored regulations that may pose an unnecessary risk of stifling privacy coins' growth. Rather, VASPs can adequately address those AML risks by maintaining an effective, risk-based program. Allowing VASPs to support privacy tokens under current, tested AML regulations strikes the appropriate policy balance between preventing money laundering and allowing beneficial, privacy-preserving technology to develop.

Introduction

Privacy-enabling cryptocurrencies, commonly known as privacy coins, are enhanced versions of early cryptocurrencies that were developed to protect the financial privacy of individuals and businesses alike. Each privacy coin leverages innovative mechanisms that provide privacy, encryption, and security to its users. Alongside their positive effects, however, these mechanisms have raised an important compliance question:

Is it possible for regulated entities to comply with anti-money laundering ("AML") obligations when supporting privacy coins?

The answer, in our view, is yes.

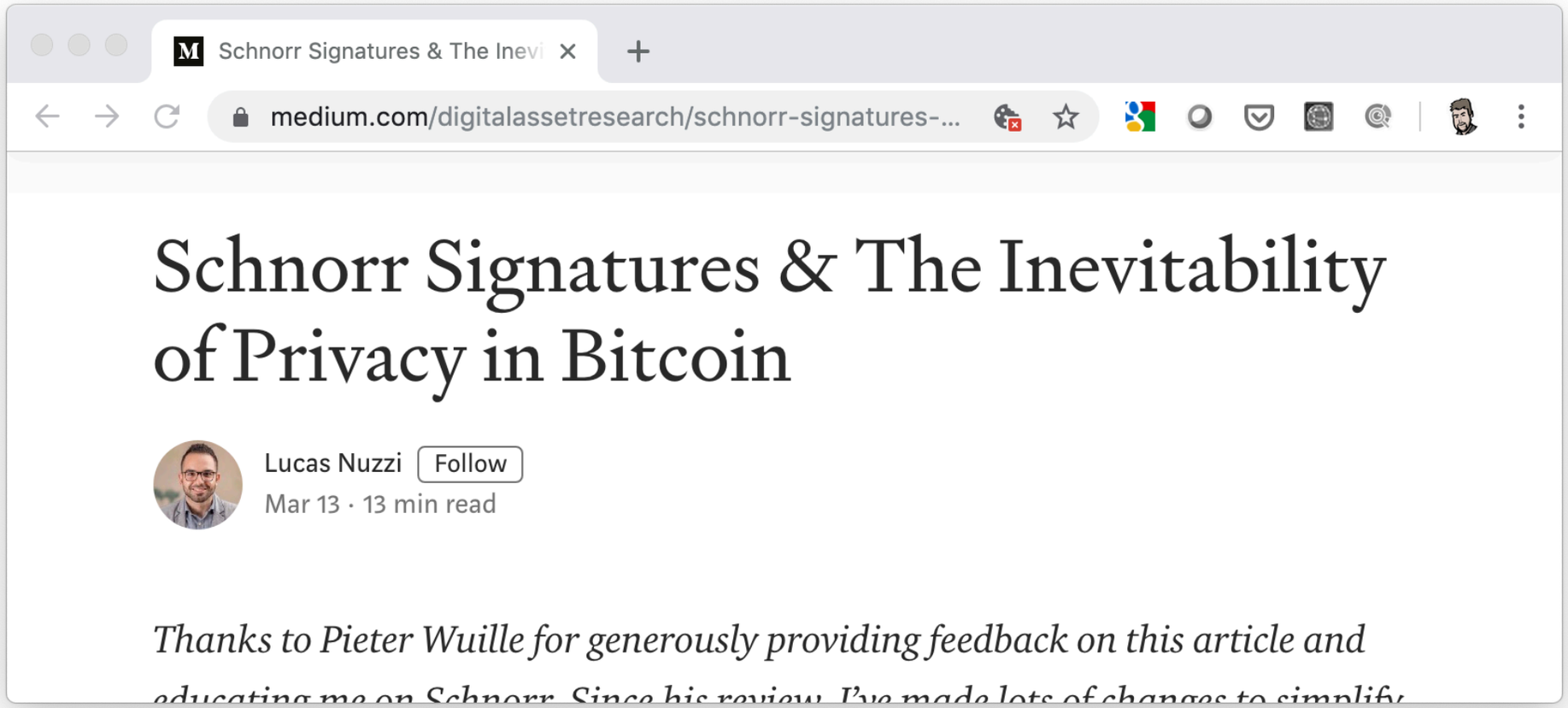
Not only do privacy coins provide public benefits that substantially outweigh their risks, existing AML regulations properly and sufficiently cover those risks, providing a proven framework for combatting money laundering and related crimes.

In reaching that conclusion, we profile several privacy coins, detail key AML regulations and intergovernmental recommendations affecting privacy coins, and explain the measures that have allowed financial institutions, intermediaries, and virtual asset service providers (collectively, "VASPs") to comply with AML obligations when facilitating privacy coin transactions. We also dispel the misconception that privacy coins are fundamentally incompatible with AML compliance, focusing on measures that have allowed VASPs to comply with AML obligations when conducting or facilitating privacy coin transactions.

What are the trade-offs?

- Blockchain traceability is not selective
  - Lack of privacy means that anyone (incl. FIS, criminals) can trace funds on the blockchain
- Traceability relies on a non-scalable approach to constructing distributed ledgers
  - Future scaling solutions are likely to be based on technologies that include privacy as a side-effect (e.g. zk-rollups)

Bitcoin is becoming a “privacy coin”…



“In time, the adoption of these technologies will make blockchain analysis obsolete…”

Ethereum is becoming a “privacy coin”…

Ethereum Upgrade Byzantium

https://cointelegraph.com/news/ethereum-upgrade-byzantium-is-live-verifies-f...

By Jon Buck SEP 21, 2017

Ethereum Upgrade Byzantium Is Live, Verifies First ZK-Snark Proof

Byzantium, the first half of the Metropolis hard fork development upgrade on the Ethereum network, has gone live. The announcement of the action was posted by Ethereum founder via Twitter:

> *Byzantium went live on Ropsten yesterday. And it just verified a zk-snark proof.*
> *https://t.co/vYm5l9BFrJ https://t.co/6XnDaIx92T*
> *— Vitalik Buterin (@VitalikButerin) September 19, 2017*

According to the Ethereum developers team, the Metropolis hard fork is due to occur late September.

In the same post, Buterin notes that Byzantium verified its first zk-snark proof.

The statement indicates that the potential for zero-knowledge proofs is real. Zero-knowledge cryptography is used as a method for verifying a ledger entry without revealing the identity of any parties.

Tezos is becoming a “privacy coin”…

zk-SNARKs and “Fine-Tuned P

medium.com/tezoscommons/zk-snar…

Tezos Insights & Updates

zk-SNARKs and “Fine-Tuned Privacy” is Coming to Tezos

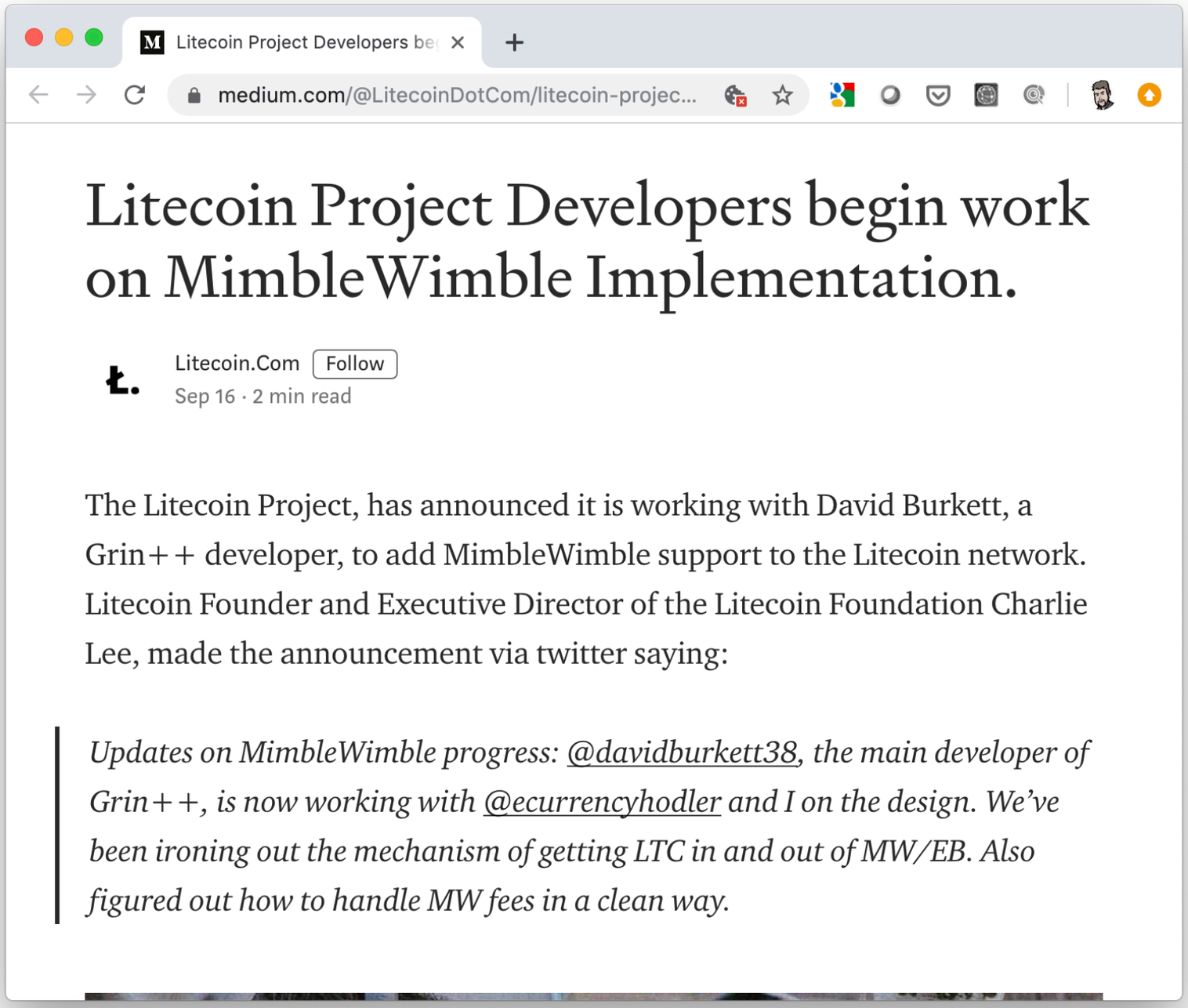
William McKenzie Follow

Oct 22 · 6 min read

zk-SNARKs is the confirmed next protocol amendment on Tezos

TLDR: zk-SNARKs is the confirmed next protocol amendment following the recent activation of the Babylon protocol amendment. Through zk-SNARKs, as enumerated on in greater detail within the post, it will provide the “fine-tuned privacy” that central and commercial banks like Bank of France can use and deploy the Tezos protocol within these entities. Additionally, in this upgrade, it will provide a new Michelson instruction that allows a smart contract to store and do a transaction on a shielded blockchain behaving like sapling.

Litecoin is becoming a “privacy coin”…



Litecoin Project Developers be

medium.com/@LitecoinDotCom/litecoin-projec...

Litecoin Project Developers begin work on MimbleWimble Implementation.

Litecoin.Com Follow

Sep 16 · 2 min read

The Litecoin Project, has announced it is working with David Burkett, a Grin++ developer, to add MimbleWimble support to the Litecoin network. Litecoin Founder and Executive Director of the Litecoin Foundation Charlie Lee, made the announcement via twitter saying:

> *Updates on MimbleWimble progress: @davidburkett38, the main developer of Grin++, is now working with @ecurrencyhodler and I on the design. We’ve been ironing out the mechanism of getting LTC in and out of MW/EB. Also figured out how to handle MW fees in a clean way.*

Conclusion

- Risk of AML / CFT involving privacy coins is not as high as perceived.
- Traceability is <u>not</u> necessary to manage AML / CFT risks.
- Blockchain technology is likely to evolve to incorporate scalability solutions that provide privacy as a side-effect.